

STM32 GPIO Data Registers

Ankit Srivastava
EE2028

Unless stated otherwise, for the second part of the module,

‘STM32 Chip’ = STM32L4S5VIT6

‘Board’ = B-L4S5I-IOT01A

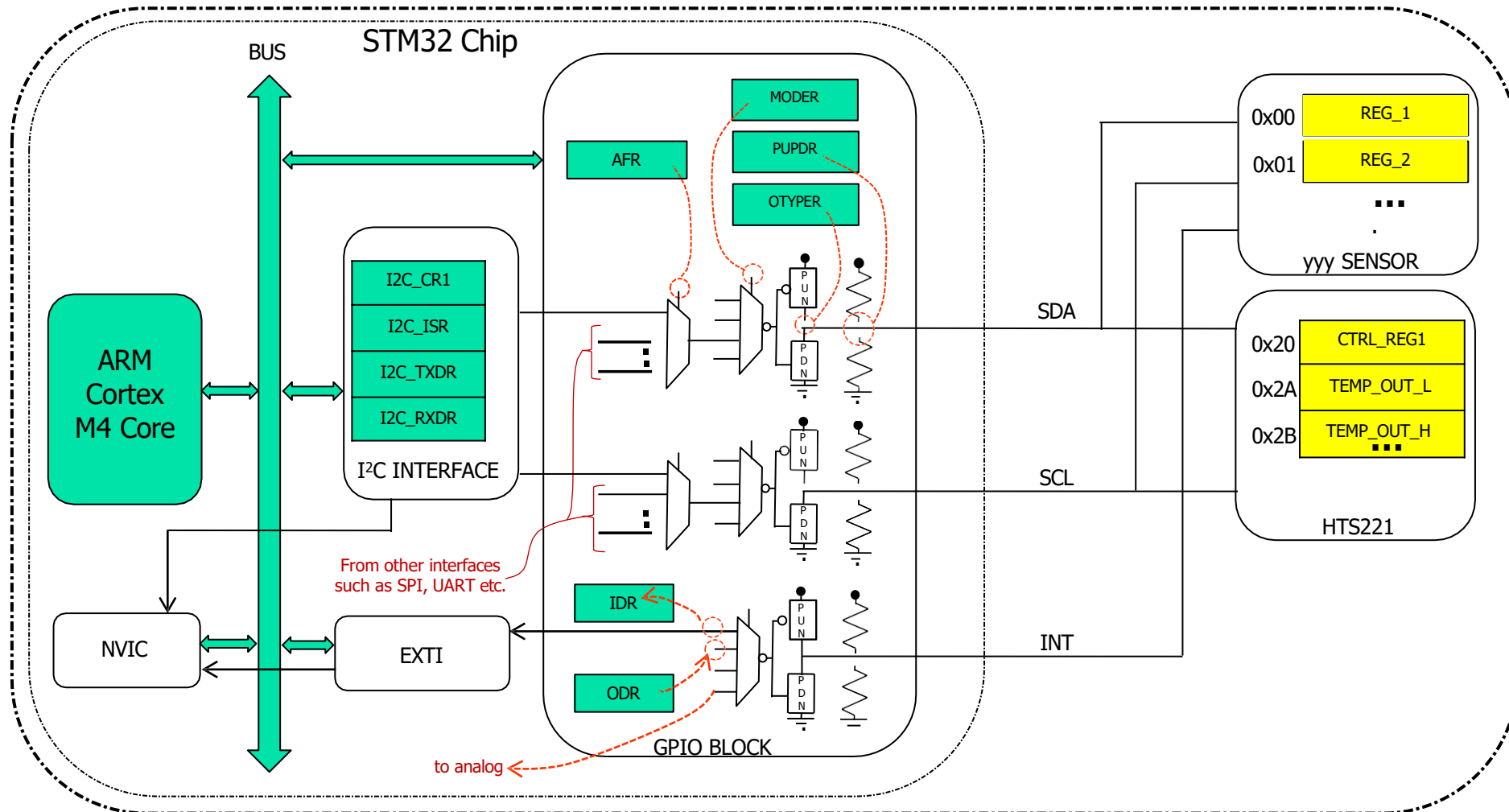
Acknowledgement: Slides adapted from Dr Rajesh Panicker



OVERVIEW

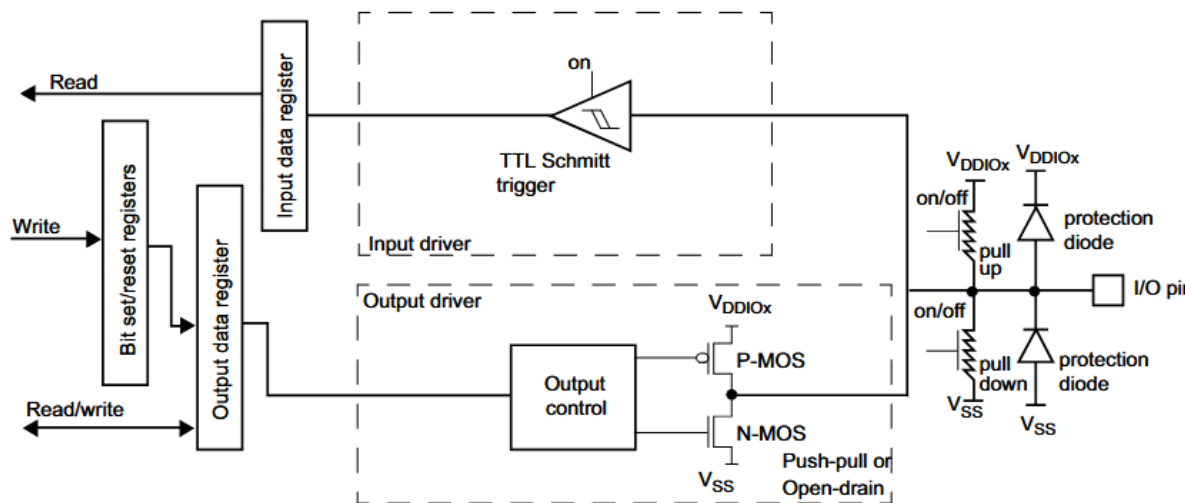
- GPIO Data Registers
 - IDR
 - ODR
 - BSRR/BRR

SYSTEM LEVEL OVERVIEW



GPIO DATA REGISTERS

- When a pin is in GPIO input mode (MODE=00), the logic level of a pin can be read from the corresponding bit in the input data register (IDR)
- When a pin is in GPIO output mode (MODE=01), the logic level of a pin will be controlled/determined by the value of the corresponding bit in the output data register (ODR)



ODR bits can also be manipulated indirectly through bit set/reset register (BSRR) and bit reset register (BRR) which are *logical* interfaces/means to affect the *physical* register ODR

Use of BSRR and BRR allows certain pins to be modified in one action (atomic), without the need to *read ODR -> modify the relevant bits -> write back to ODR*

GPIO DATA REGISTERS: GPIOx_IDR & GPIOx_ODR

- GPIOx_IDR (Address offset = 0x10) : Input Data Register (read-only)
- GPIOx_IDR ID_y contains the logic level read from port **x** pin **y** (**y**=15:0). Bits 31:16 of this register is reserved
- E.g. To know the input level at Port **C** Pin **13** (**x**=C, **y**=13): check bit value at GPIO**C**_IDR [13]

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID15	ID14	ID13	ID12	ID11	ID10	ID9	ID8	ID7	ID6	ID5	ID4	ID3	ID2	ID1	ID0
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r

- GPIOx_ODR (Address offset = 0x14) : Output Data Register
- GPIOx_ODR OD_y contains the logic level output on port **x** pin **y** (**y**=15:0). Bits 31:16 of this register is reserved
- E.g., To output 1 from Port **B** Pin **14** (**x**=B, **y**=14): Set GPIOB_ODR [14] = 1 and keep the other bits unchanged
- ODR is read-write. We can read back and verify the output when the pin is a GPIO output. Reading ODR will not give the pin logic level in GPIO input mode

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OD15	OD14	OD13	OD12	OD11	OD10	OD9	OD8	OD7	OD6	OD5	OD4	OD3	OD2	OD1	OD0
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

GPIO DATA REGISTERS: GPIOx_BRR & GPIOx_BSRR

- GPIOx_BRR (Address offset = 0x28) : Bit Reset Register
- GPIOx_BRR BRy, for port x pin y (y = 15:0) works as below. These bits are write-only, a read to these bits returns the value 0x0000. Bits 31:16 of this register is reserved

Set = make a bit 1
Reset = make a bit 0

- 0**: no action on the corresponding ODy bit
- 1**: Resets the corresponding ODy bit

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BR15	BR14	BR13	BR12	BR11	BR10	BR9	BR8	BR7	BR6	BR5	BR4	BR3	BR2	BR1	BR0
w	w	w	w	w	w	w	w	w	w	w	w	w	w	w	w

- GPIOx_BSRR (Address offset = 0x18) : Bit Set/Reset Register (write-only)
- GPIOx_BSRR BSy, for port x pin y (y = 15:0) works as below

- 0**: no action on the corresponding ODy bit
- 1**: Sets the corresponding ODy bit

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
BR15	BR14	BR13	BR12	BR11	BR10	BR9	BR8	BR7	BR6	BR5	BR4	BR3	BR2	BR1	BR0
w	w	w	w	w	w	w	w	w	w	w	w	w	w	w	w
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BS15	BS14	BS13	BS12	BS11	BS10	BS9	BS8	BS7	BS6	BS5	BS4	BS3	BS2	BS1	BS0
w	w	w	w	w	w	w	w	w	w	w	w	w	w	w	w

- GPIOx_BSRR BRy, for port x pin y (y = 15:0) works as below
- 0**: no action on the corresponding ODy bit
- 1**: Resets the corresponding ODy bit

BRy of BSRR is intended to be used only when there is a need to set some pins and reset some other pins *at the same instant*

GPIO DATA REGISTERS: GPIOx_BRR & GPIOx_BSRR

GPIOx_BSRR

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
BR15	BR14	BR13	BR12	BR11	BR10	BR9	BR8	BR7	BR6	BR5	BR4	BR3	BR2	BR1	BR0
												1	1		
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BS15	BS14	BS13	BS12	BS11	BS10	BS9	BS8	BS7	BS6	BS5	BS4	BS3	BS2	BS1	BS0
											1		1		

GPIOx_ODR

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OD15	OD14	OD13	OD12	OD11	OD10	OD9	OD8	OD7	OD6	OD5	OD4	OD3	OD2	OD1	OD0
											1	0	1		



CONCLUSION

- GPIO Data Registers
 - IDR
 - ODR
 - BSRR/BRR